

WORLD HAPPINESS REPORT 2021-ANALYSIS

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INTRODUCTION

In recent years, policymakers and global organizations have increasingly recognized that economic growth alone does not equate to societal wellbeing. To better understand the components of human happiness and explore their relationships, I conducted an exploratory data analysis based on the **World Happiness Report 2021**, which evaluates 149 countries using six primary variables: **GDP per capita, social support, healthy life expectancy, freedom to make life choices, generosity, and perceptions of corruption.**

This report was completed as part of my self-directed learning in Excel, aiming to improve data handling, statistical analysis, and visualization skills. While the conclusions here are preliminary and not exhaustive, they reflect my capacity to clean, analyze, and interpret real-world data in a structured and critical way. All insights are intended as a starting point for further exploration and refinement using more advanced tools such as R, Python, or SQL in the future.

METHODOLOGY

The main goals of this project were:

- To explore linear relationships between happiness and various socio-economic factors,
- To assess regional differences in key variables and rankings,
- To test how alternative composite scores could alter country rankings,
- To simulate hypothetical policy changes (e.g., increasing GDP in African countries) and analyze their potential effects.

Methodological steps included:

1. Cleaning the dataset (e.g., correcting decimal separators),
2. Applying correlation analysis using =CORREL() to compute Pearson coefficients,
3. Using pivot tables and formulas (AVERAGEIFS, RANK, IFS) to aggregate and compare data,
4. Creating visualizations (line plots, comparisons, heatmaps) to support interpretation,
5. Designing a custom linear regression model for predicting happiness scores based on all six parameters.

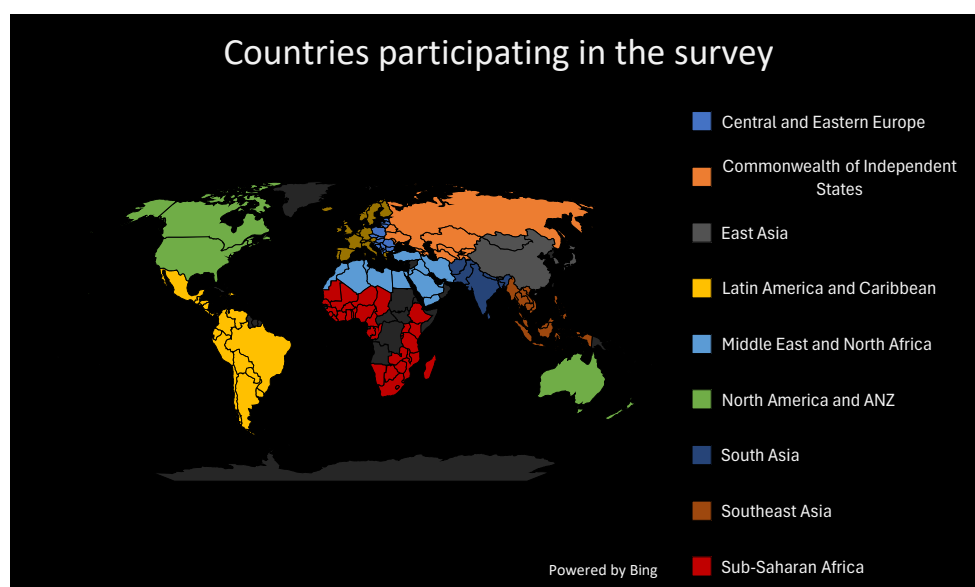
DATA ANALYSIS AND VISUAL REPRESENTATION

The happiness scores and rankings in the **World Happiness Report 2021** are based on data from the Gallup World Poll, covering 149 countries. The ladder score—interpreted as the happiness score—is calculated using six factors: **GDP per capita, social support, healthy life expectancy, freedom to make life choices, generosity, and perceptions of corruption.**

However, happiness is a complex and multidimensional concept. In addition to socio-economic indicators, it is influenced by cultural, genetic, psychological, and familial factors. Therefore, any analysis of such data must be approached critically and cautiously, with an understanding that simplified models may not fully reflect reality.

Data Cleaning

Upon importing the dataset into Excel, decimal commas were incorrectly interpreted as thousands separators. For example, a life expectancy value of "71.600" was read as seventy-one thousand six hundred. To correct this, variables such as **GDP**, **life expectancy**, and **happiness score** were divided by 1,000. Fortunately, the dataset was complete, requiring minimal additional cleaning.



Graph1. Visual representation of countries involved in the survey.

Correlation Analysis

To explore relationships between happiness and the six variables, the **Pearson correlation coefficient (r)** was calculated using Excel's =CORREL() function for each pair. The values are interpreted as follows:

Factor	Pearson coefficient r (between factor and happiness score)	Interpretation
GDP	0,79	Strong positive correlation
Social support	0,76	Strong positive correlation
Healthy life expectancy	0,77	Strong positive correlation
Freedom to make life choices	0,61	Strong positive correlation
Generosity	-0,02	No correlation
Perceptions of corruption	-0,42	Weak negative correlation

*Table1. his analysis indicates that **GDP**, **social support**, and **life expectancy** are most strongly associated with happiness. **Generosity** shows no linear correlation, and **corruption** exhibits a weak negative relationship.*

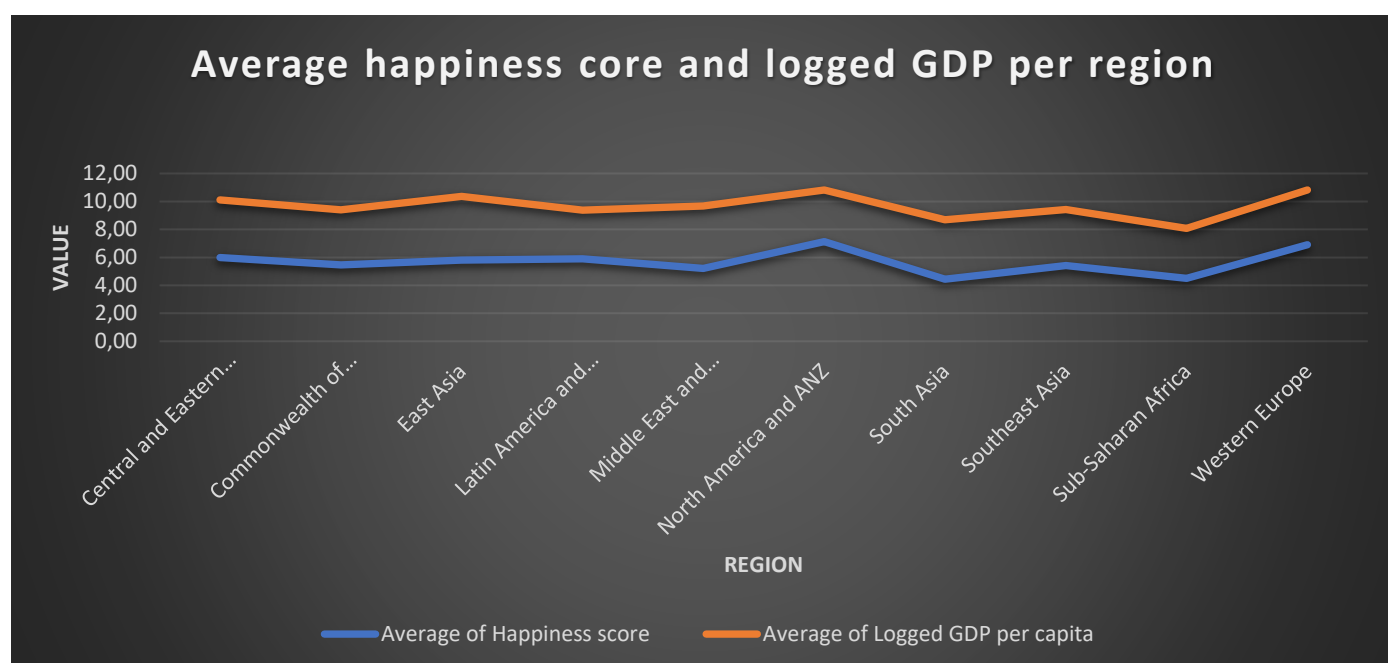
Regional Comparison

Using a pivot table, I calculated average happiness scores, logged GDP per capita, and country counts by region.

I could get the same values by using functions =AVERAGEIFS() and =COUNTIFS().

Row Labels	Average of Happiness score	Average of Logged GDP per capita	Number of countries
Central and Eastern Europe	5,985	10,109	17
Commonwealth of Independent States	5,467	9,402	12
East Asia	5,810	10,368	6
Latin America and Caribbean	5,908	9,370	20
Middle East and North Africa	5,220	9,666	17
North America and ANZ	7,129	10,810	4
South Asia	4,442	8,683	7
Southeast Asia	5,408	9,421	9
Sub-Saharan Africa	4,494	8,075	36
Western Europe	6,915	10,823	21
Grand Total	5,533	9,432	149

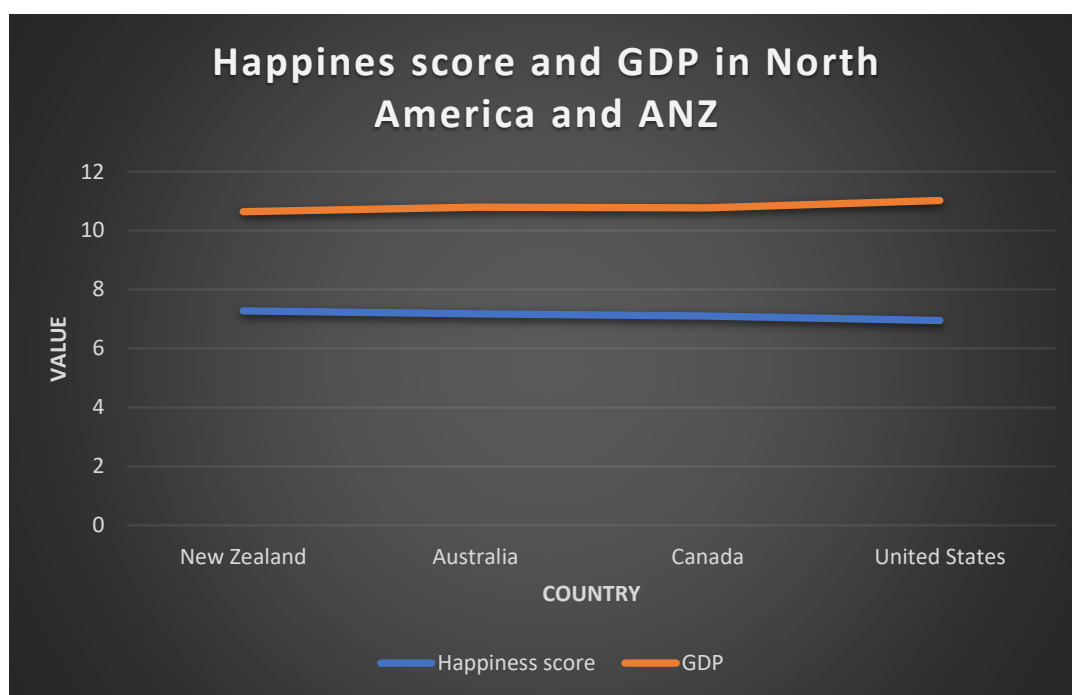
Table2. Western Europe has the highest average logged GDP but second highest happiness score while North America and ANZ has the other way around. Sub-Saharan Africa has the highest amount of countries in the region (36), which gives credibility to their average scores. In North America and ANZ a single country with high values can drastically influence the average happiness and logged GDP scores.



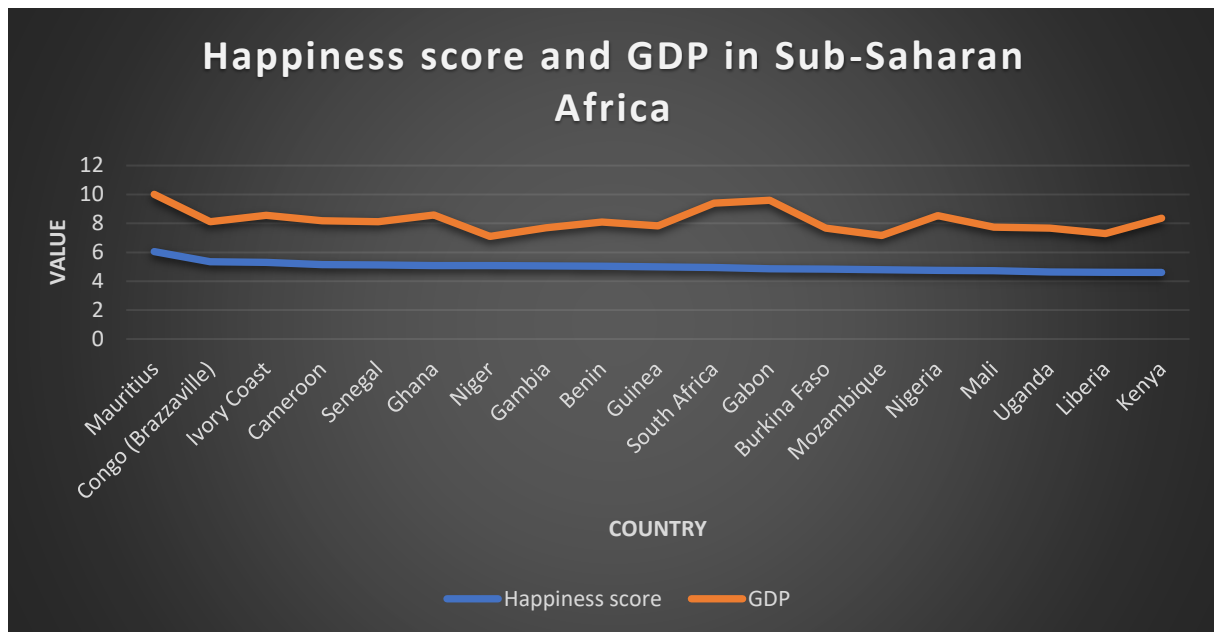
Graph2. We can clearly see that both happiness score and logged GDP curves have similar shapes which can be explained with a high degree of linear correlation. Keep in mind that these values are only the averages.

Intra-Regional Correlations

I further examined **North America/ANZ** and **Sub-Saharan Africa** individually to assess within-region relationships between GDP and happiness. In **North America/ANZ**, the strong correlation may be misleading due to the limited number of countries. In **Sub-Saharan Africa**, although more data points are available, the relationship appears non-linear and inconsistent across countries.



Graph3. Curves of happiness score and GDP do have similar shapes, which could indicate linear correlation. But because of too few countries, this conclusion can be deceiving as shown on graph4.



Graph4. Curves of happiness score and GDP are sometimes similar while other time very contradicting, which hints on non-linear correlation. Because the graph includes sufficient data pool, I trust this graph more than graph3.

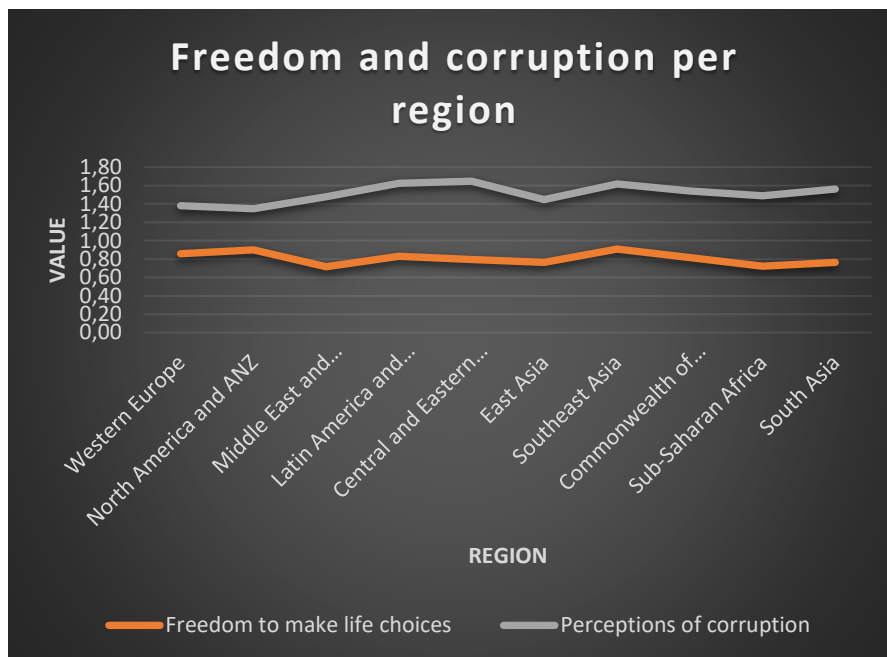
Freedom and Corruption

To explore the relationship between **freedom to make life choices** and **perceptions of corruption**, I calculated a correlation of $r = -0.40$, indicating a weak inverse relationship. However, when visualized by region, this trend is not consistently visible—freedom and corruption do not always move in opposite directions. Regional averages show that **Southeast Asia** reports the highest degree of freedom, while **Sub-Saharan Africa** and **Middle East & North Africa** report the lowest. Using a pivot table I simply calculated average freedom and average corruption per region (instead of using functions =AVERAGEIF(), =UNIQUE() and =COUNTIFS()).

	Number of countries	Freedom to make life choices	Perceptions of corruption
Western Europe	21	0,86	0,52
North America and ANZ	4	0,90	0,45
Middle East and North Africa	17	0,72	0,76
Latin America and Caribbean	20	0,83	0,79
Central and Eastern Europe	17	0,80	0,85
East Asia	6	0,76	0,68
Southeast Asia	9	0,91	0,71
Commonwealth of Independent States	12	0,82	0,73
Sub-Saharan Africa	36	0,72	0,77
South Asia	7	0,77	0,80

Table3. People of Southeast Asia have the highest average degree of freedom, while for people from Africa and Middle East have the lowest score. Perception of corruption is lowest in North America and ANZ but highest in Central and Eastern Europe. Kaggle.com has a bad

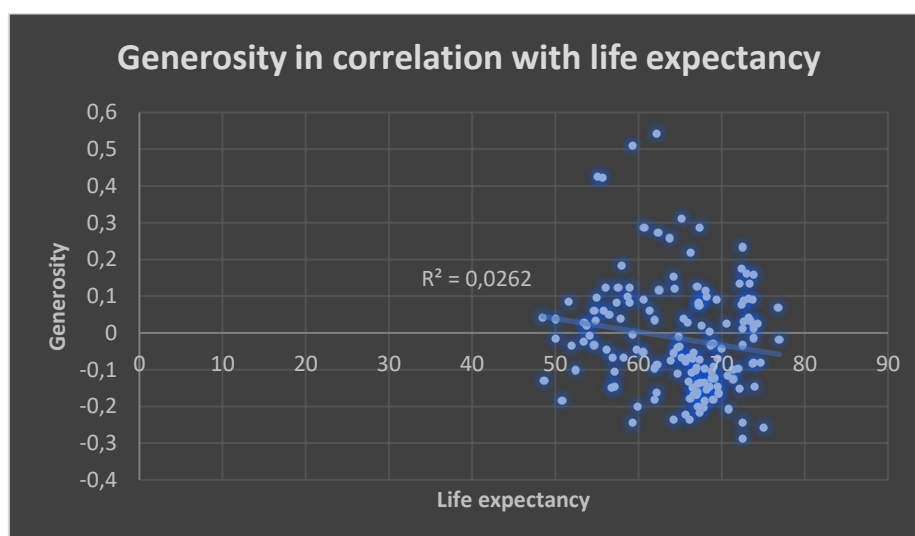
habbit of not explaining the meaning of each variable, how it was calculated and which scale was used.



Graph5. According to Pearson coefficient curves of both values should be opposite. When one value rises the other should fall and vice versa. But here that trend is barely noticeable, unless you specifically look for it. I could compare it within different regions but this is not the main focus of this report. More freedom to make life choices doesn't mean there is less corruption.

Generosity vs. Life Expectancy

The correlation between **healthy life expectancy** and **generosity** was weak and negative ($r = -0.16$), with a very low R^2 value of 0.026. This indicates no meaningful linear relationship between these two variables. The scatterplot confirms high variability and no clear trend.



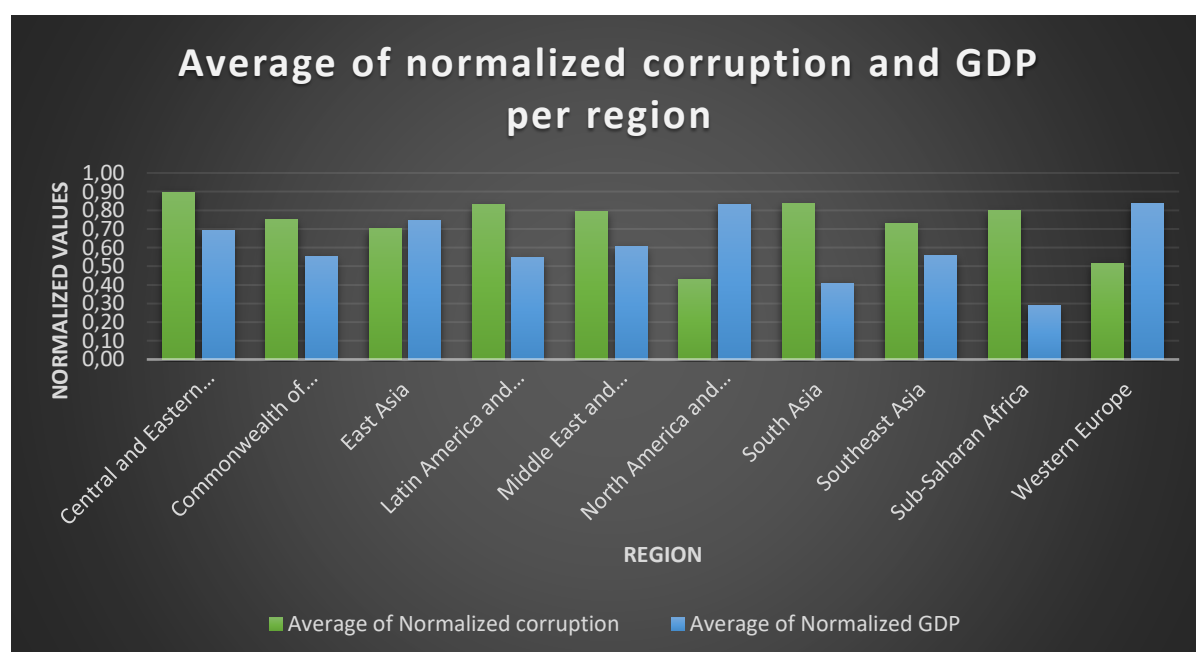
Graph6. No significant correlation between healthy life expectancy and generosity.

Normalized Comparison: GDP and Corruption

In my next task I wanted to visually present the differences between corruption and GDP per region. Often the problem with different variables is the difference between their orders of magnitude. Corruption is measured on a scale from 0 to 1, while GDP is on a logarithmic scale ranging roughly from 5 to 10. That's why I first normalized GDP and corruption by using the following formula: $N = \frac{value - min}{max - min}$ and using functions =MIN() and =MAX(). Value in this is the value we want to normalize; min is the smallest value in the variable (GDP or corruption) and max is the highest value in the variable. After that I calculated the difference between them and highlighted them with Conditional Formatting. I could also use function =ABS() to show the absolute difference.

Regions	Average of Normalized corruption	Average of Normalized GDP	Difference
Central and Eastern Europe	0,90	0,69	-0,20
Commonwealth of Independent States	0,75	0,55	-0,20
East Asia	0,70	0,74	0,04
Latin America and Caribbean	0,83	0,55	-0,28
Middle East and North Africa	0,79	0,60	-0,19
North America and ANZ	0,43	0,83	0,40
South Asia	0,83	0,41	-0,43
Southeast Asia	0,73	0,56	-0,18
Sub-Saharan Africa	0,80	0,29	-0,51
Western Europe	0,51	0,84	0,32

Table4. The biggest difference between logged GDP and corruption is Sub-Saharan Africa, where corruption is relatively higher than in North America and ANZ. Minor differences between values are due to rounding.



Graph7. Average of normalized corruption and GDP per region.

Custom Ranking: "Wellbeing Score"

To evaluate how rankings shift when factors are combined differently, I calculated a "Wellbeing Score":

Wellbeing score = GDP + Social support + Freedom – Corruption

Countries were then ranked based on both their happiness (ladder) scores and this composite metric. Only four countries—**Slovenia (29)**, **Sweden (7)**, **Canada (14)**, and **Cyprus (39)**—maintained the same position under both rankings, underscoring how sensitive rankings are to methodology and weights.

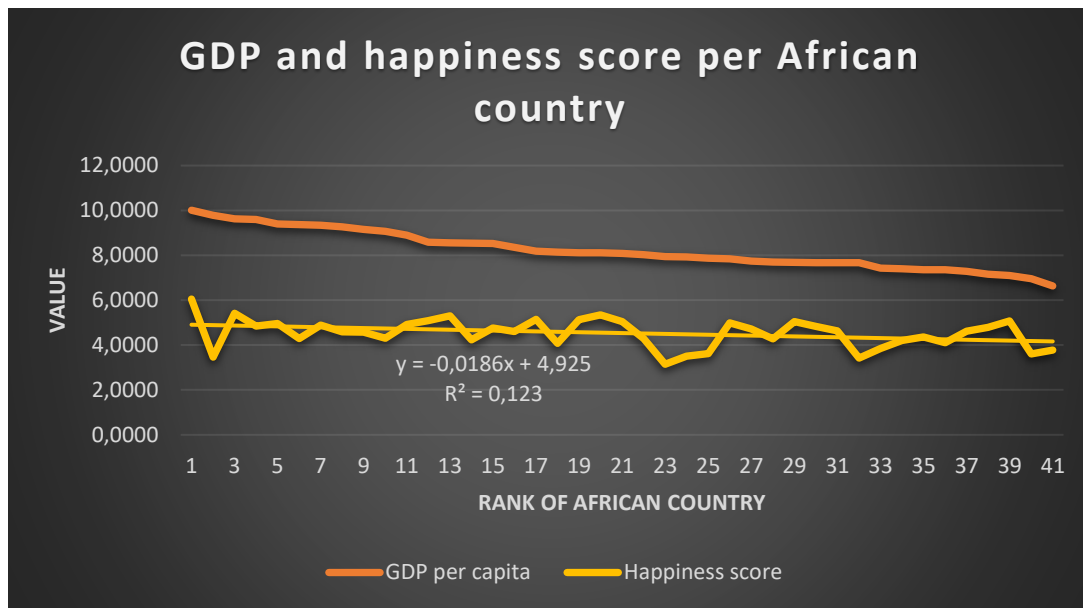
Rank	Wellbeing score	Ladder score
1	Singapore	Finland
2	Luxembourg	Denmark
3	Ireland	Switzerland
4	Norway	Iceland
5	Switzerland	Netherlands
6	Denmark	Norway
7	Sweden	Sweden
8	Finland	Luxembourg
9	Netherlands	New Zealand
10	New Zealand	Austria

Table 5. Almost all the countries are from both scores are from Europe.

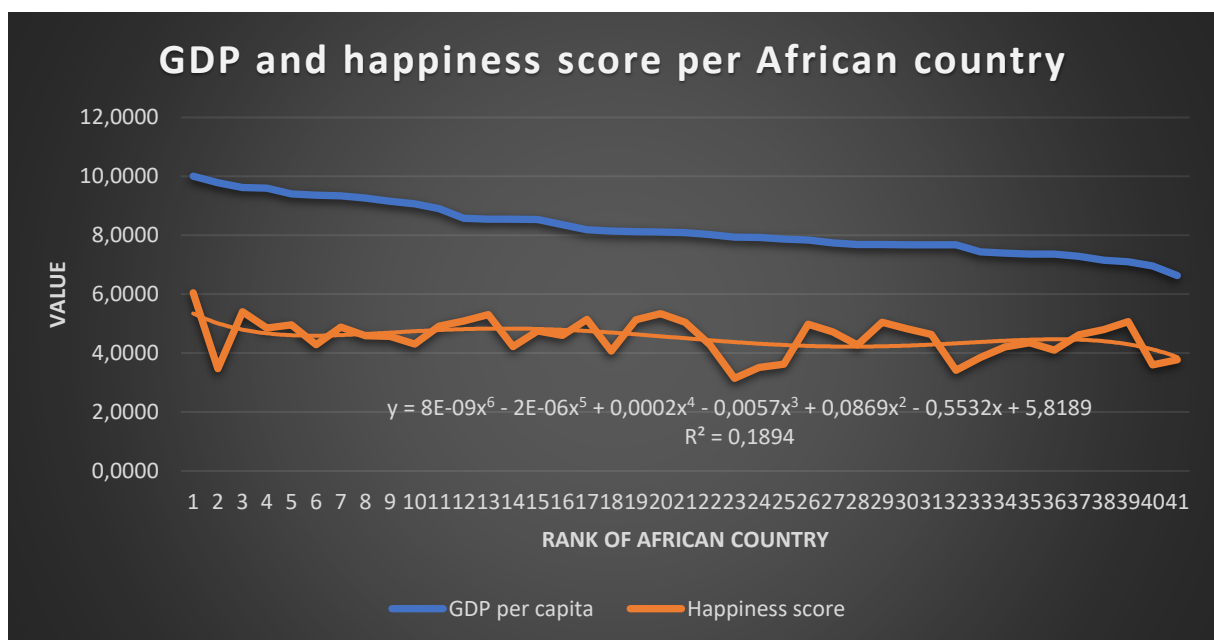
Simulation: GDP Increase in Africa

To explore the hypothetical impact of economic growth, I simulated a **20% increase in GDP** for African countries (including Sub-Saharan Africa and parts of MENA by filtering). Various models (linear, polynomial, exponential) were tested to predict the corresponding happiness score changes.

The best-fitting model was **polynomial**, yet even it had an R^2 value too low for reliable prediction. This reinforces the idea that happiness in Africa is not solely GDP-driven and is likely shaped by numerous social and political factors.



Graph8. Both curves and R^2 indicate non-linear correlation between GDP and happiness in Africa.



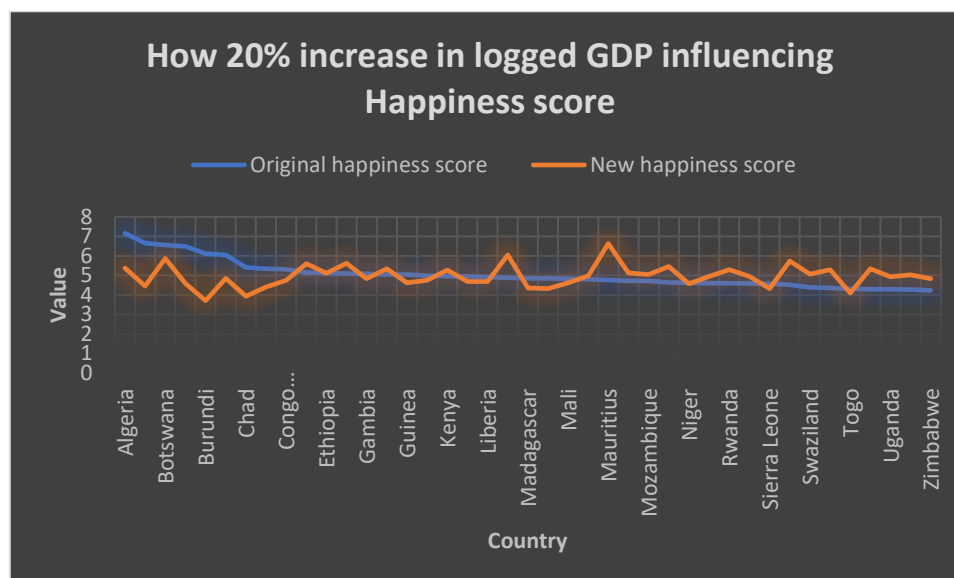
Graph9. Using polynomial trend line R^2 is higher but still too low for anything more concrete and reliable.

Regression Model

Using all six factors, I derived the following **multiple linear regression model**:

$$\text{happiness} = -2,2372 + 0,2795 * (\text{Logged GDP per capita}) + 2,4762 * (\text{Social support}) + 0,0303 * (\text{Healthy life expectancy}) + 2,0105 * (\text{Freedom to make life choices}) + 0,3644 * (\text{Generosity}) - 0,6051 * (\text{Perceptions of corruption})$$

*This model predicts happiness with an average error of **0.41 points**, ranging from **0.0067 to 1.85**. However, even with all variables included, predictions vary—supporting the notion that happiness is not fully captured by this limited dataset.*



Graph10. Increase in GDP doesn't necessarily mean an increase in happiness. This can be explained by inaccurate formula, data or complexity of happiness.

Using a custom legend and function =IFS() I created a heat map of perceptions of corruption per regions with Conditional Formatting:

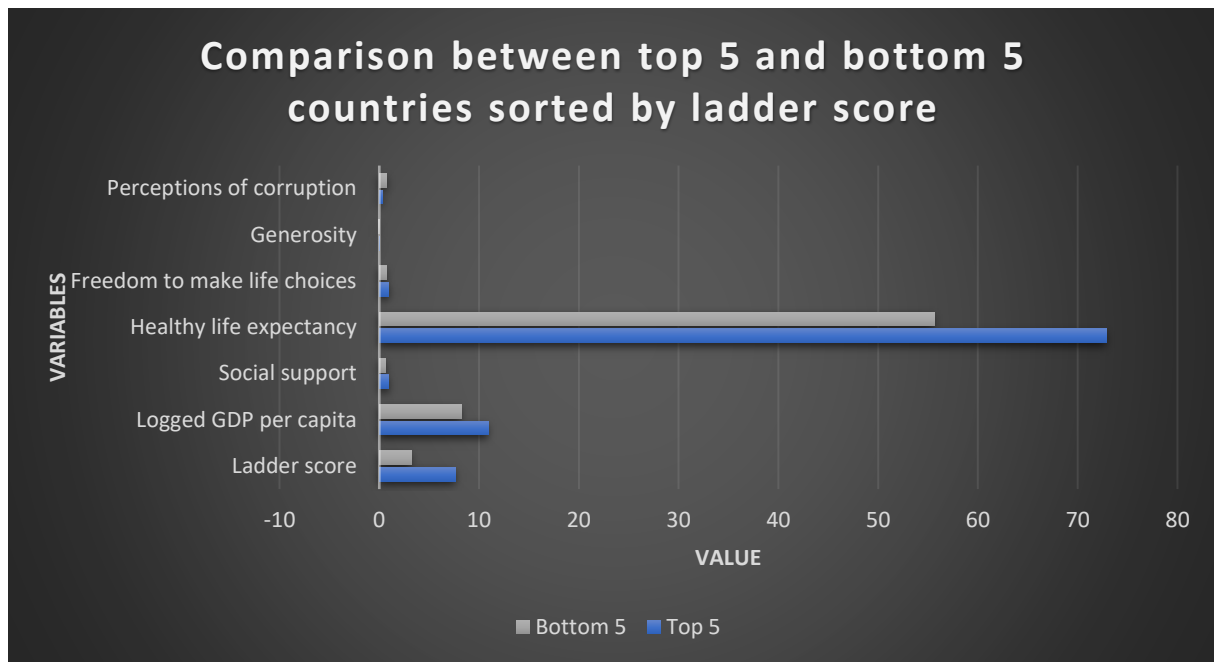
High corruption: 0,082-0,500

Medium corruption: 0,501-0,700

Low corruption: 0,701-0,939

Southeast Asia	Singapore	0,082	High corruption
Southeast Asia	Thailand	0,895	Low corruption
Southeast Asia	Philippines	0,742	Low corruption
Southeast Asia	Vietnam	0,796	Low corruption
Southeast Asia	Malaysia	0,839	Low corruption
Southeast Asia	Indonesia	0,867	Low corruption
Southeast Asia	Laos	0,658	Medium corruption
Southeast Asia	Cambodia	0,843	Low corruption
Southeast Asia	Myanmar	0,66	Medium corruption

Table6. Perceived corruption in Southeast Asia.



Graph11. Because data has different scales the presentation is lacking. This could be fixed with normalizing them.

CONCLUSION

This project demonstrates how key socio-economic indicators relate to national happiness, as measured by the **World Happiness Report 2021**. While six variables (GDP per capita, social support, life expectancy, freedom, generosity, and perceptions of corruption) form the backbone of the dataset, this analysis confirms that **no single factor** can fully explain happiness. Nevertheless, several insights emerged:

- **GDP per capita** showed the strongest positive linear correlation with happiness ($r = 0.79$), followed closely by **social support** and **life expectancy**.
- **Generosity** had no significant linear correlation, and **freedom** did not always correlate with lower perceived corruption.
- **Sub-Saharan Africa** had the largest data pool ($n = 36$), but also the lowest average GDP and happiness score.
- **North America and ANZ** showed the **largest positive difference** between GDP and corruption (i.e., high GDP, low perceived corruption).
- A custom regression model using all six factors (formula below) predicted happiness scores with an average error of **0.41 points**:

$$\text{happiness} = -2,2372 + 0,2795 * (\text{Logged GDP per capita}) + 2,4762 * (\text{Social support}) + 0,0303 * (\text{Healthy life expectancy}) + 2,0105 * (\text{Freedom to make life choices}) + 0,3644 * (\text{Generosity}) - 0,6051 * (\text{Perceptions of corruption})$$

- Regional disparities are striking: despite economic growth, **African countries** did not see a corresponding rise in happiness. A 20% simulated GDP increase in these countries failed to produce a proportional improvement in happiness due to weak model fit (R^2 too low), suggesting **nonlinear or multi-factorial influences**.
- Only four countries retained the same position when comparing ladder rankings vs. custom "Wellbeing Score" rankings, underlining how weighting of variables can greatly shift perceptions of wellbeing.

In conclusion, this analysis supports the idea that **wellbeing is multifactorial**, and its interpretation requires caution. While economic strength matters, **freedom, fairness, and social cohesion** play equally vital roles in promoting national happiness. Future analyses should expand the dataset with additional indicators (e.g., inequality, education, political stability) and use advanced modeling techniques to improve predictive accuracy.